

EMBEDDING AGENTIC TRACES FOR EFFICIENT EVALUATION

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ABSTRACT

Evaluating a new agentic system on an existing benchmark is often necessary to understand its behavior before deployment. For modern agentic evaluation frameworks, generating and grading a trace for every task can be prohibitively expensive – often hundreds to thousands of dollars for a single system. In practice, traces from previously-evaluated systems are often cached – creating a potential opportunity to use this additional information to decrease the number of tasks required to accurately evaluate a new system. In this paper, we extend an approach for predicting benchmark performance from cached responses to the agentic setting, where the central challenge is representing the trace itself. We first demonstrate that off-the-shelf semantic embeddings of traces are dominated by evaluation-irrelevant confounders – harness, model family, and exact configuration – rather than by the content of the work performed, and that predictions based on them do not improve with additional probe tasks. We then introduce QUBRIC, a query-specific-rubric transformation that converts off-the-shelf embedding functions into content-based, confounder-invariant trace representations. Empirically, on SWE-bench Verified (107 systems), combining the resulting trace geometry with item response theory predicts full-benchmark resolve rates within 0.066 mean absolute error from a single probe task and 0.032 from twenty, achieving the best or near-best error at every probe budget under a conservative leave-one-LLM-out protocol. We conclude by quantifying the corresponding evaluation savings and releasing the reference geometry, which is constructed from public archives for approximately \$26.

1 INTRODUCTION

The dominant paradigm of generative model use is increasingly agentic: systems that plan, invoke tools, observe results, and iterate over long horizons rather than emit a single response. Evaluation is a necessary part of understanding an agentic system’s capabilities and limitations before deployment. As these systems become more capable and are trusted with more consequential work, the tasks on which they must be evaluated become correspondingly complex. For example, recent agentic benchmarks such as SWE-bench (Jimenez et al., 2024; OpenAI, 2024), Terminal-Bench (Terminal-Bench Team, 2026), ARC-AGI (Chollet, 2019), and multi-benchmark agent suites (Kapoor et al., 2026) require generating up to hundreds of thousands of tokens per task¹ and evaluate each system on roughly one hundred to two thousand tasks. At reported per-task costs of \$0.13–\$4 (an aggregate \$1.84 per rollout across 21,730 rollouts in Kapoor et al., 2026), fully evaluating a single system costs roughly \$100–\$2,000, and multi-benchmark sweeps cost tens of thousands of dollars. At the same time, agentic systems are sensitive to changes in system prompts, retrieval databases, scaffold parameters, and underlying model versions – making it infeasible to fully evaluate each new variant. The result is systematic under-evaluation: most checkpoints, ablations, and configurations are never scored (Kapoor et al., 2024).

These trends motivate query-efficient alternatives to full benchmark evaluation. In other evaluation settings, several such alternatives exist: benchmark compression and anchor-item selection

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¹In the SWE-bench Verified leaderboard archive we measure a median of $\sim 40K$ tokens per trace, with a heavy tail exceeding 700K tokens.

(Vivek et al., 2024; Maia Polo et al., 2024), item response theory and adaptive testing (Kipnis et al., 2025; Truong et al., 2025), and performance prediction from observational records (Ruan et al., 2024; Zhuang et al., 2025). One line of work uses off-the-shelf embedding functions to construct low-dimensional representations of the models themselves and casts benchmark prediction as an inference problem in the resulting space (Duderstadt et al., 2023; Helm et al., 2025; 2026). These methods require only black-box access, require no additional labeled data, and leverage the response caches that evaluation ecosystems already produce. Extending them to the agentic setting requires an embedding function for traces. Unfortunately, off-the-shelf embedding functions do not yet exist for agentic traces. Further, we find that semantic embedding functions applied to traces without modification are not appropriate for efficient evaluation: the resulting representations are dominated by evaluation-irrelevant *confounders* – harness, model family, and exact configuration – rather than by the content of the work performed. In this paper, we introduce QUBRIC (*query-specific rubrics*), a transformation that converts off-the-shelf embedding functions into content-based, confounder-invariant trace representations suitable for efficient evaluation.

Contribution. We make two primary contributions.

1. **Methodological:** We introduce QUBRIC, which extracts and embeds trace content according to a query-specific rubric. Whereas off-the-shelf trace representations recover harness, model family, and exact configuration at well above chance while remaining near chance on outcome, QUBRIC representations collapse the confounder axes to approximately chance while retaining task and behavioral signal, robustly across seven embedding functions.
2. **Empirical:** We demonstrate that QUBRIC-based methods are query-efficient. Combined with item-response-theoretic estimators, they achieve the best or near-best mean absolute error at every probe budget and are significantly better where graded evidence is scarce – e.g., 0.066 MAE from a single probe task. We release the reference geometry, code, and judged-trace corpus – constructed from public archives for approximately \$26 – to support future work on trace-based evaluation.

1.1 PROBLEM STATEMENT

Given an agentic system f , we consider the problem of predicting its score on a benchmark $Q^* = \{q_1, \dots, q_M\}$. For our purposes, a system $f \in \mathcal{F}$ is a random mapping from a query space $\mathcal{Q}$ to a trace space $\mathcal{X}$: executing f on query q yields a trace $f(q)$, a long, semi-structured log of the system’s actions, observations, and outputs. Let $s : \mathcal{X} \rightarrow \{0, 1\}$ denote the harness’s trace-level scoring function and $y : \mathcal{F} \times 2^{Q^*} \rightarrow [0, 1]$ denote the benchmark scoring function, $y(f, Q) = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{E}[s(f(q))]$. We refer to $y(f, Q^*)$ as y when the context is clear.

As described above, full evaluation of f on all queries may be infeasible. Instead, we assume access to previously-evaluated (system, score) pairs $(f_1, y_1), \dots, (f_n, y_n)$, along with each reference system’s trace and trace-level score for every query. Such reference caches exist publicly at scale.² Given this additional information, our goal is to estimate y with $m \ll M$ queries. That is, we seek to identify an estimate $\hat{y}$ that predicts the full benchmark score of f from its traces (and, optionally, its trace-level scores) on a subset $Q \in 2^{Q^*}$ and the information available from the previously-evaluated systems.

Two aspects distinguish this setting from the short-response setting of Helm et al. (2026). First, the observation associated with each query is not a short response but a trace of 10^4 – 10^6 tokens in one of many mutually incompatible harness formats. Whether and how traces can be embedded usefully is the central question of this paper. Second, the reference population is confounded: reference systems that share the target’s underlying language model or harness permit trivially accurate prediction via score-copying. We therefore evaluate under a leave-one-LLM-out protocol – references sharing the target’s underlying language model are excluded – and verify robustness under a leave-one-harness-out protocol. The query subset Q may be fixed, drawn at random, or selected adaptively per target; we report all three regimes.

²The SWE-bench Verified leaderboard archive contains traces and outcomes for over one hundred systems; we estimate it embodies \$50K–\$100K of agent compute contributed by submitters.

2 BACKGROUND & RELATED WORK

Agents, traces, and evaluations. Agentic benchmarks pair long-horizon tasks with automated grading harnesses. Examples include SWE-bench and its verified subset for repository-level software issues (Jimenez et al., 2024; OpenAI, 2024), executed by scaffolds such as SWE-agent (Yang et al., 2024) and OpenHands (Wang et al., 2025); Terminal-Bench for command-line tasks (Terminal-Bench Team, 2026); and suites that aggregate many such settings (Kapoor et al., 2026). The traces these systems produce have recently become objects of study in their own right. Mehtiyev & Assunção (2026) analyze leaderboard trajectories via symbolic action alphabets, Shu et al. (2026) document per-scaffold format fragmentation, Cemri et al. (2025) taxonomize failure modes, Zhuge et al. (2025) propose trajectory-level judging, and Xiao et al. (2025) study trajectory compression. This literature produces taxonomies and statistics rather than reusable vector representations. Notably, its own attribution results imply that naive trace representations are dominated by confounders: coding-agent trajectories are attributable to scaffold and model at 85.7% accuracy against an 11% baseline (Oderinwale, 2026), and model-level fingerprints survive paraphrasing and summarization (Sun et al., 2025; Wu et al., 2026). Benchmark-integrity findings – memorization inflating resolve rates (Liang et al., 2025) and mislabeled gold patches (Yu et al., 2025) – further motivate representations that capture how a system works rather than relying on the pass/fail bit alone.

Efficient evaluations. Several lines of work aim to reduce the computational cost of benchmark evaluation. Vivek et al. (2024) select “anchor points” from clusters of cross-model predictions, Maia Polo et al. (2024) use item response theory to construct benchmark subsets of approximately 1% of the original size, Kipnis et al. (2025) compress across benchmarks jointly, and Truong et al. (2025) amortize item calibration for adaptive testing. These techniques consume only correctness patterns – a few bits per probe – which imposes a resolution floor at extreme budgets (Yauney et al., 2026). Efficiency work specific to agents is nascent: Ndzomga (2026) observe that subset selection degrades under scaffold shift, and Ge et al. (2026) extend psychometric models with task features while still consuming correctness rather than trace content. Closest to our work is the Data Kernel Perspective Space (DKPS) lineage (Duderstadt et al., 2023; Helm et al., 2025; Acharyya et al., 2024), and in particular query-efficient evaluation from cached responses (Helm et al., 2026), which predicts a new model’s benchmark score by comparing its responses on a small query subset to cached responses from previously-evaluated models. We extend this framework to the agentic setting, where each observation is a trace rather than a short response and where the confounded reference population requires an invariant representation and a conservative reference protocol.

Embeddings of non-standard objects. General-purpose text embedding functions (Reimers & Gurevych, 2019) fix one notion of similarity at training time, and no single embedding can represent all relevance relations (Weller et al., 2025). Instruction-conditioned variants (Su et al., 2023) honor only coarse, trained-in task tags, and conditional similarity (Deshpande et al., 2023) formalizes similarity-under-a-criterion at the sentence scale. A related family of methods embeds LLM-generated text in place of the original object: answers to an instruction (Peng et al., 2024), answer vectors to question batteries (Benara et al., 2024), and generated descriptions (Ravfogel et al., 2024). Embeddings of interaction data exist for short dialogues (Liu et al., 2022) and for reinforcement learning trajectories (Ge et al., 2025) – the latter observing that trajectory embeddings encode agent “style,” precisely the confounder our setting must remove. QUBRIC differs from these approaches in that it re-represents very long, heterogeneous logs at inference time without training, includes an explicit invariance mechanism (per-query consensus centering), and is evaluated with respect to confounder invariance rather than assuming it.

3 QUBRIC

4 EXPERIMENTS

5 DISCUSSION

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